CIE AS PHYSICS (9702) • QUESTION PAPER

3.3 Linear Momentum & its Conservation

Syllabus Section: 3. Dynamics **Paper Type:** Multiple Choice (Paper 1) **Total:** 34 Questions (34 Marks)

Question 1 Easy [1 mark]

What is the principle of conservation of momentum?

A force is equal to the rate of change of momentum

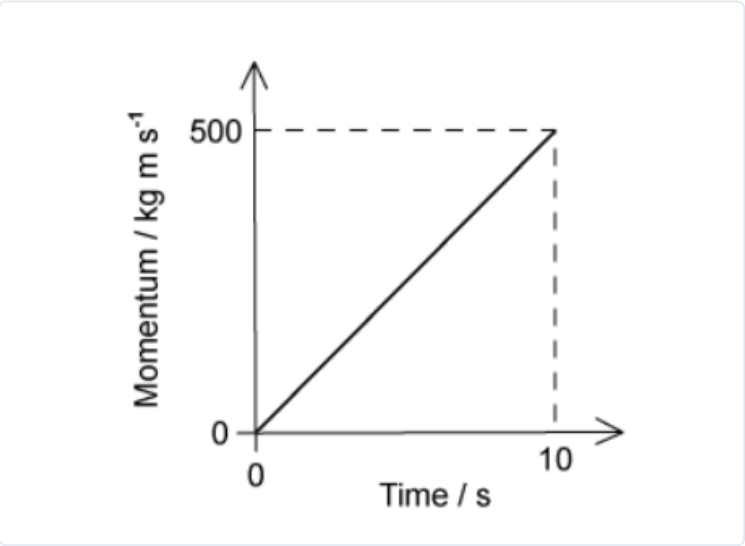
B momentum is the product of mass and velocity

C the total momentum of two bodies after collision is equal to their total momentum before collision

D the total momentum of a system remains constant provided no external force acts on it

Question 2 Easy [1 mark]

The graph shows how the momentum of a motorcycle changes with time.



What is the resultant force on the motorcycle?

A 50 N

B 500 N

C 2500 N

D 5000 N

Question 3 Easy [1 mark]

Two bodies travelling in a straight line collide in a perfectly elastic collision. Which of the following statements must be correct?

A the initial speed of one body will be the same as the final speed of the other body

B the relative speed of approach between the two bodies equals their relative speed of separation

C the total momentum is conserved but the total kinetic energy will be reduced

D one of the bodies will be stationary at one instant

Question 4 Easy [1 mark]

Which row correctly states whether momentum and kinetic energy are conserved in an inelastic collision in which there are no external forces?

	Momentum	Kinetic energy
A	conserved	conserved
B	not conserved	conserved
C	conserved	not conserved
D	not conserved	not conserved

A

B

C

D

Question 5 Easy [1 mark]

Which quantity has the same base units as momentum?

A density × volume × velocity

B density × energy

C pressure × area

D weight ÷ area

Question 6 Easy [1 mark]

Which of the following is a statement of the principle of conservation of momentum?

A in an elastic collision momentum is constant

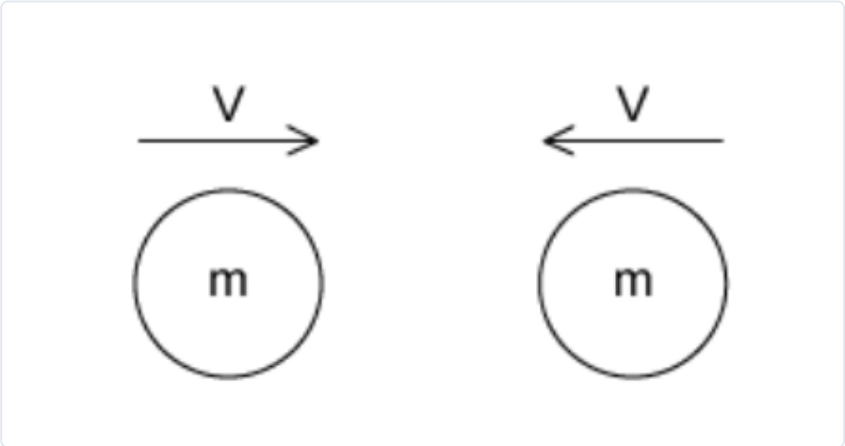
B momentum is the product of mass and velocity

C the force acting on a body is proportional to its rate of change of momentum

D the momentum of an isolated system is constant

Question 7 Easy [1 mark]

Two similar spheres, each of mass *m* and travelling with speed *v*, are moving towards each other



The spheres have a head-on elastic collision

Which statement is correct?

A the spheres stick together on impact

B the total kinetic energy after impact is mv^2

C the total kinetic energy before impact is zero

D the total momentum before impact is 2*mv*

Question 8 Easy [1 mark]

Trolley X, moving along a horizontal frictionless track, collides with a stationary trolley Y. The two trolleys become attached and move off together.

Which statement about this interaction is correct?

A some of the kinetic energy of trolley X is changed to momentum in the collision

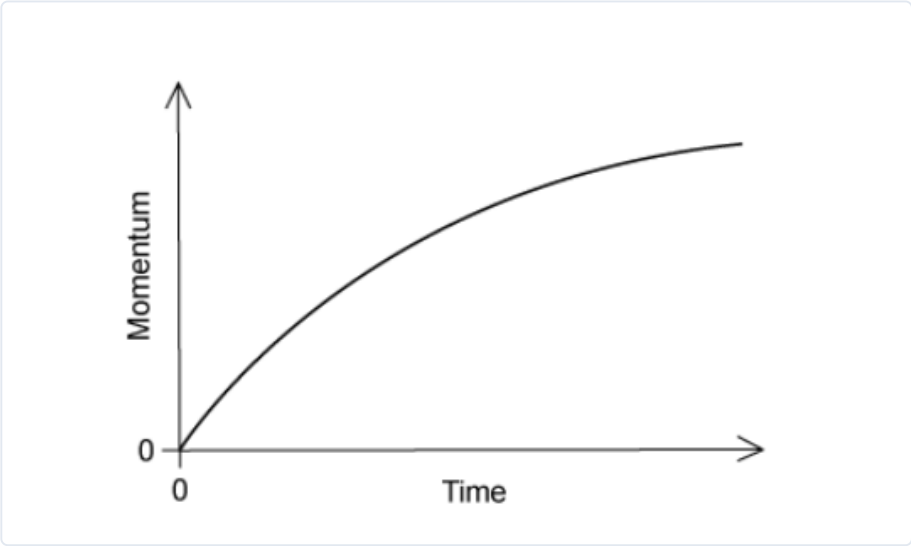
B some of the momentum of trolley X is changed to kinetic energy in the collision

C trolley X loses some of its momentum as heat in the collision

D trolley X shares its momentum with trolley Y but some kinetic energy is lost

Question 9 Easy [1 mark]

A car accelerates from rest. The graph shows the momentum of the car plotted against time.



What is the meaning of the gradient of the graph at a particular time?

A the velocity of the car

B the resultant force on the car

C the kinetic energy of the car

D the rate of change of kinetic energy of the car

Question 10 Easy [1 mark]

Which statement about a ball that strikes a tennis racket and rebounds is **always** correct?

A total kinetic energy of the ball is conserved

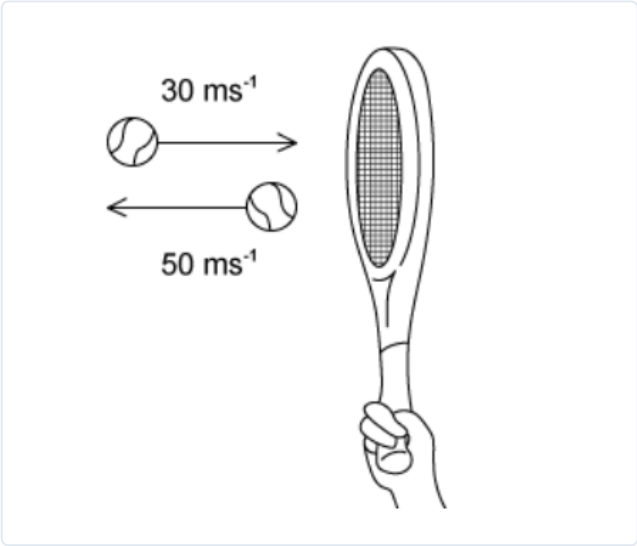
B total kinetic energy of the system is conserved

C total momentum of the ball is conserved

D total momentum of the system is conserved

Question 11 Medium [1 mark]

A tennis ball of mass 60 g is struck by a tennis racket. The velocity of the ball changes as shown



What is the magnitude of the change in momentum of the ball?

A 1.2 kg m s^{-1}

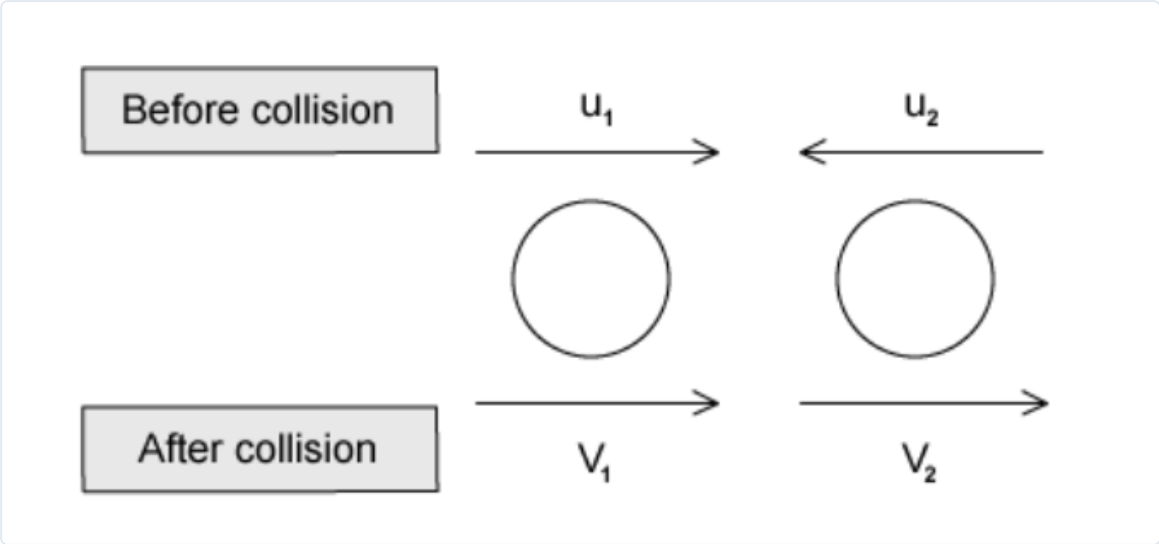
B 4.8 kg m s^{-1}

C 1200 kg m s^{-1}

D 4800 kg m s^{-1}

Question 12 Medium [1 mark]

Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2 before collision, and v_1 and v_2 after collision, in the directions shown below



Which equation is correct if the collision is perfectly elastic?

A $u_1 - u_2 = v_2 + v_1$

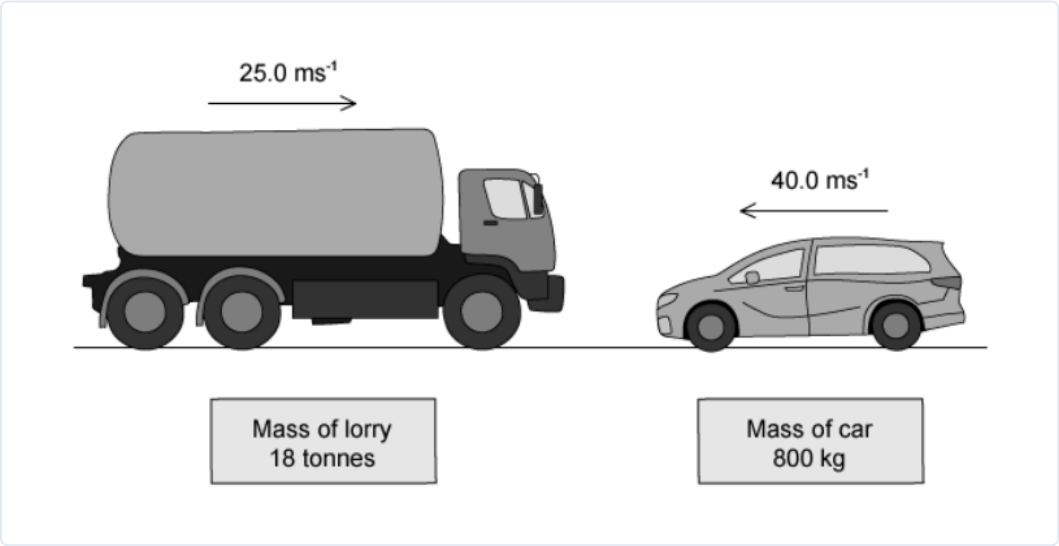
B $u_1 - u_2 = v_2 - v_1$

C $u_1 + u_2 = v_2 - v_1$

D $u_1 + u_2 = v_2 + v_1$

Question 13 Medium [1 mark]

A lorry of mass 18 tones is travelling at 25.0 $m \quad s^{-1}$



A car of mass 800 kg is travelling at 40.0 $m \quad s^{-1}$ towards the lorry.

What is the magnitude of the total momentum?

A 282 kN s

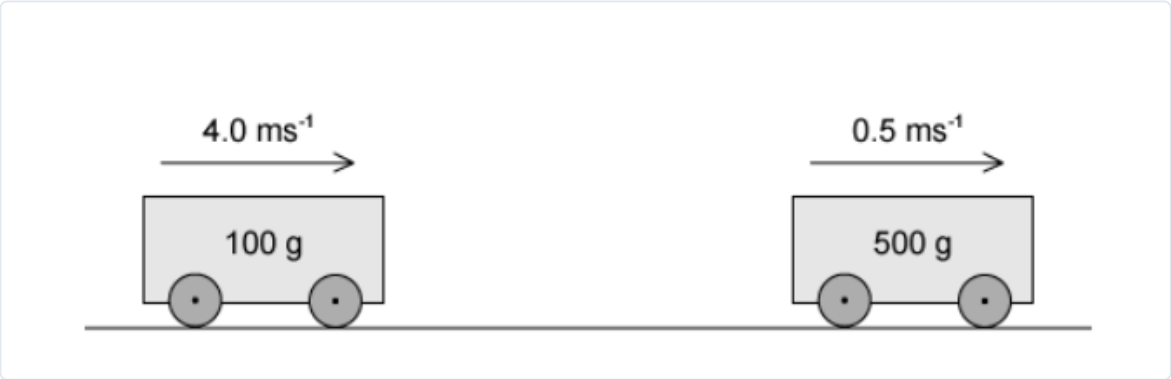
B 418 kN s

C 482 kN s

D 1222 kN s

Question 14 Medium [1 mark]

The diagram shows the masses and velocities of two trolleys about to collide



After the impact they move off together

What is the total kinetic energy of the trolleys after the collision?

A 6.0 J

B 3.7 J

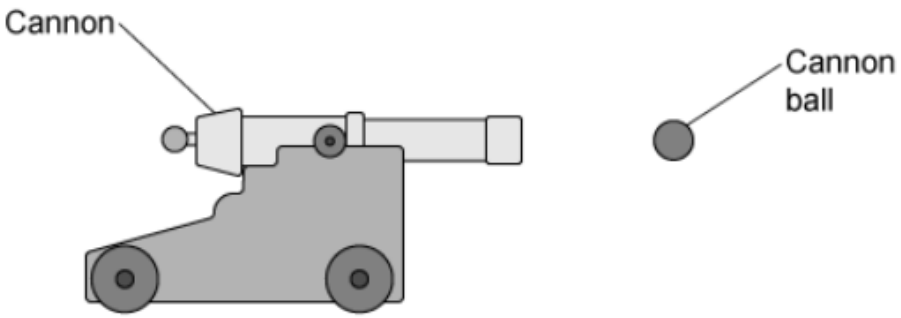
C 0.35 J

D 0.02 J

Question 15 Medium [1 mark]

The diagram shows a cannon ball fired from a cannon

The mass of the cannon is 1600 kg and the mass of the cannon ball is 20 kg



The recoil velocity of the cannon is 5 m s^{-1} horizontally

What is the horizontal velocity of the cannon ball?

A 100 m s^{-1}

B 320 m s^{-1}

C 400 m s^{-1}

D 8000 m s^{-1}

Question 16 Medium [1 mark]

A ball of mass 2 kg travelling at ½ m s^{-1} strikes a ball of mass 3 kg travelling at ⅓ m s^{-1}. Both balls are moving along the same straight line as shown



After collision, both balls move at the same velocity *v*

What is the magnitude of the velocity *v*?

A $\frac{1}{5}$ m s^{-1}

B $\frac{2}{5}$ m s^{-1}

C $\frac{3}{2}$ m s^{-1}

D $\frac{5}{2}$ m s^{-1}

Question 17 Medium [1 mark]

A molecule of mass m travelling at speed v hits a wall in a direction perpendicular to the wall. The collision is elastic.

What are the changes in the momentum and in the kinetic energy of the molecule caused by the collision?

	Change in momentum	Change in kinetic energy
A	0	0
B	mv	mv^2
C	mv^2	0
D	$2mv$	0

A

B

C

D

Question 18 Medium [1 mark]

A moving object strikes a stationary object. The collision is inelastic. The objects move off together.

Which row shows the possible values of total momentum and total kinetic energy for the system before and after the collision?

	Total momentum before collision / kg m s^{-1}	Total momentum after collision / kg m s^{-1}	Total kinetic energy before collision / J	Total kinetic energy after collision / J
A	6	6	90	30
B	6	6	30	90
C	6	2	90	30
D	6	6	90	90

A

B

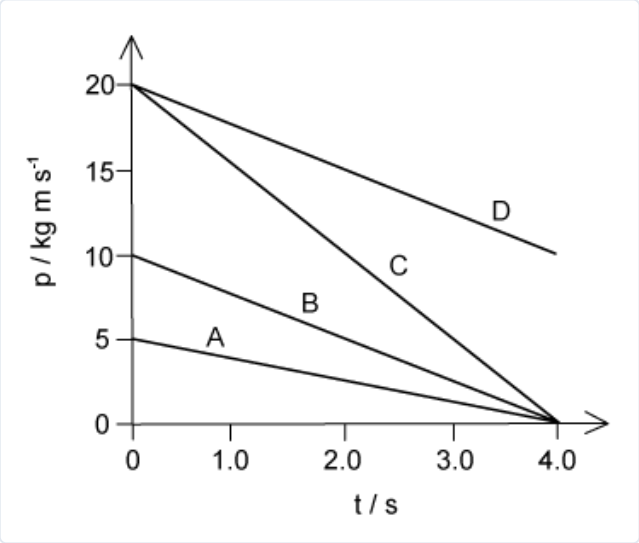
C

D

Question 19 Medium [1 mark]

A resultant force of 5 N acts on a body for a time of 4.0 s

Which graph could show the variation with time *t* of the momentum *p* of the body?



A

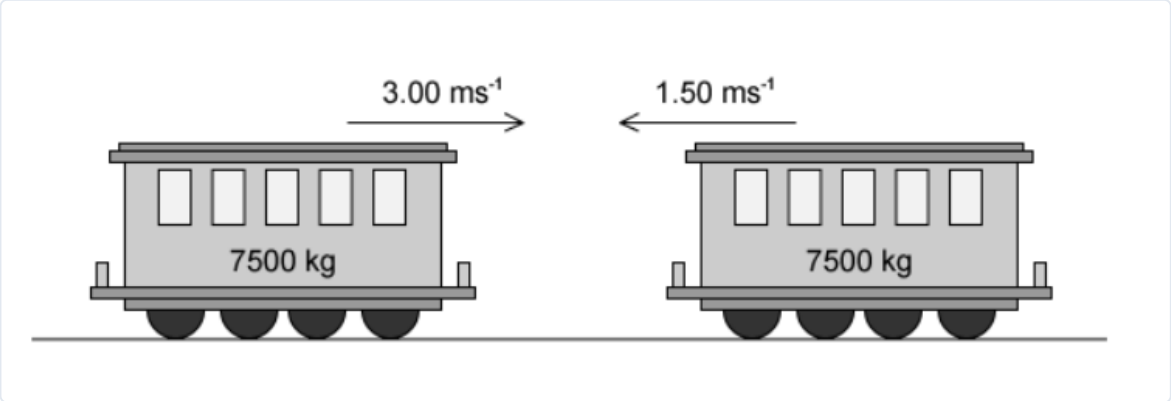
B

C

D

Question 20 Medium [1 mark]

Two train carriages each of mass 7500 kg roll toward one another on a level track. One is travelling at 3.00 m s^{-1} and the other at 1.50 m s^{-1} as shown.



They collide and join together.

What is the kinetic energy lost during the collision?

A 4200 J

B 7500 J

C 11 250 J

D 38 000 J

Question 21 Medium [1 mark]

A spacecraft of mass 1000 kg and a fuel module of mass m are initially travelling together at 13 m s^{-1} in a straight line.

The fuel module separates from the rocket and continues to travel along the same line at 5 m s^{-1} while the spacecraft continues to travel along the same line at 17 m s^{-1}.

What is the mass m of the fuel module?

A 300 kg

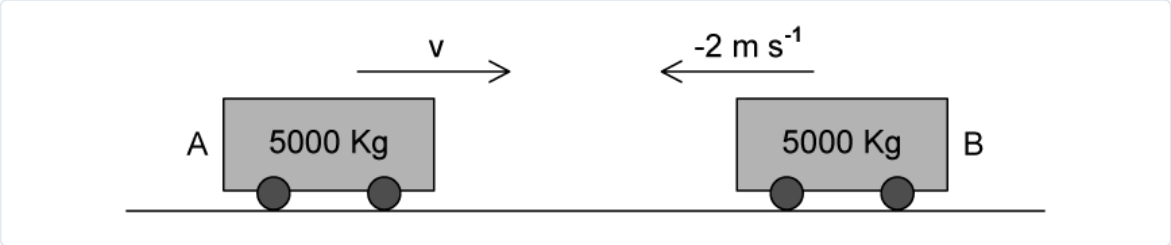
B 500 kg

C 800 kg

D 1000 kg

Question 22 Medium [1 mark]

Two trolleys, A and B, each have a mass of 5000 kg. Trolley A travels at speed v towards trolley B. Trolley B travels at 2.0 m s^{-1} in the opposite direction, as shown.



The trolleys collide and join together with a combined kinetic energy of 1250 J.

What is the velocity of trolley A before the collision?

A -1.0 m s^{-1}

B 3.0 m s^{-1}

C 3.4 m s^{-1}

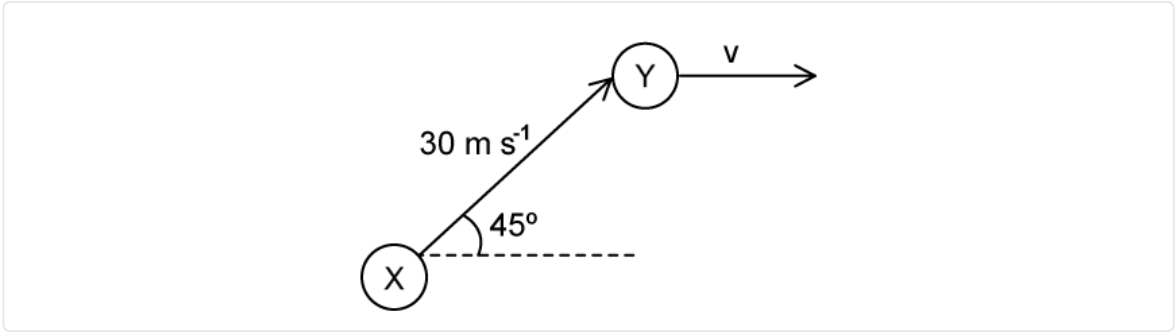
D 4.0 m s^{-1}

Question 23 Medium [1 mark]

Two identical billiard balls of mass m are placed on a smooth horizontal surface.

Ball Y is initially at rest. Ball X approaches ball Y with a speed of 30 m s^{-1} at an angle of 45° to the direction shown.

The balls collide perfectly elastically. After the collision, ball Y moves with speed v, and ball X moves at right angles to the direction of motion of ball Y.



What is the speed v of ball Y after the collision?

A 21 m s^{-1}

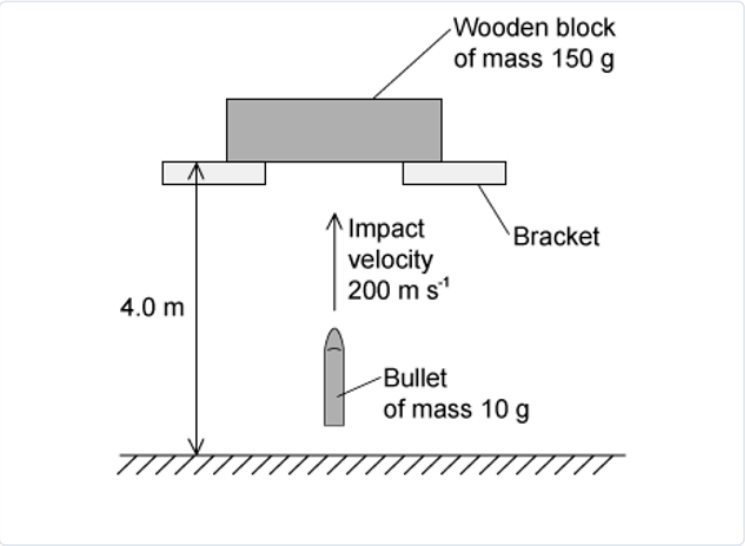
B 26 m s^{-1}

C 30 m s^{-1}

D 39 m s^{-1}

Question 24 Hard [1 mark]

A wooden block is freely supported on brackets at a height of 4.0 m above the ground, as shown



A bullet of mass 10.0 g is shot vertically upwards into the wooden block of mass 150 g. It embeds itself in the block. The impact causes the block to rise above its supporting brackets.

The bullet hits the block with a velocity of 200 m s^{-1}. How far above the ground will the block be at the maximum height of its path?

A 8.0 m

B 10.0 m

C 12.0 m

D 14.0 m

Question 25 Hard [1 mark]

Two railway trucks of masses *m* and 5 *m* move towards each other in opposite directions with speeds 3 *u* and *u* respectively. These trucks collide and stick together.

What is the speed of the trucks after the collision?

A $\frac{u}{6}$

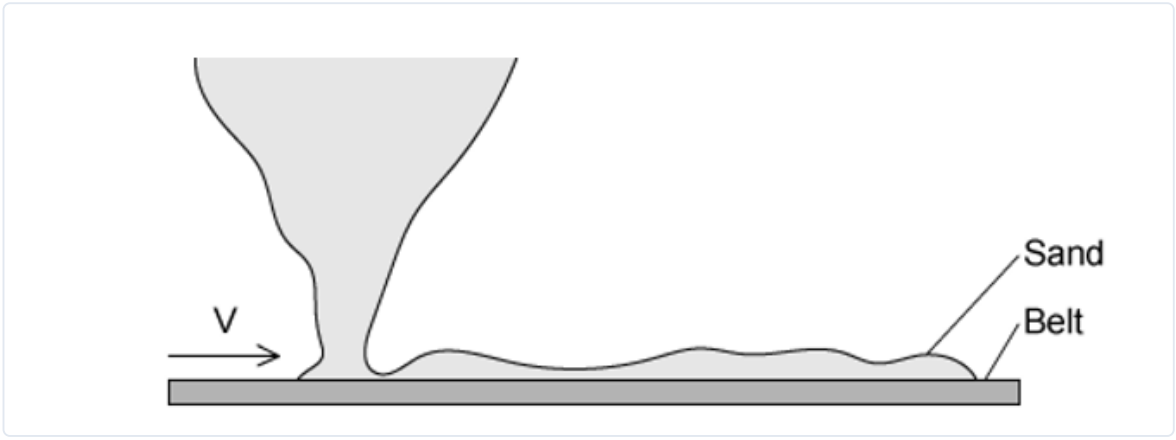
B $\frac{3u}{6}$

C $\frac{u}{3}$

D $\frac{5u}{3}$

Question 26 Hard [1 mark]

Sand falls vertically on a conveyor belt at a rate of m kg s^{-1}



In order to keep the belt moving at constant speed v, the horizontal force that must be exerted is equal to

A mv

B $\frac{1}{2} mv$

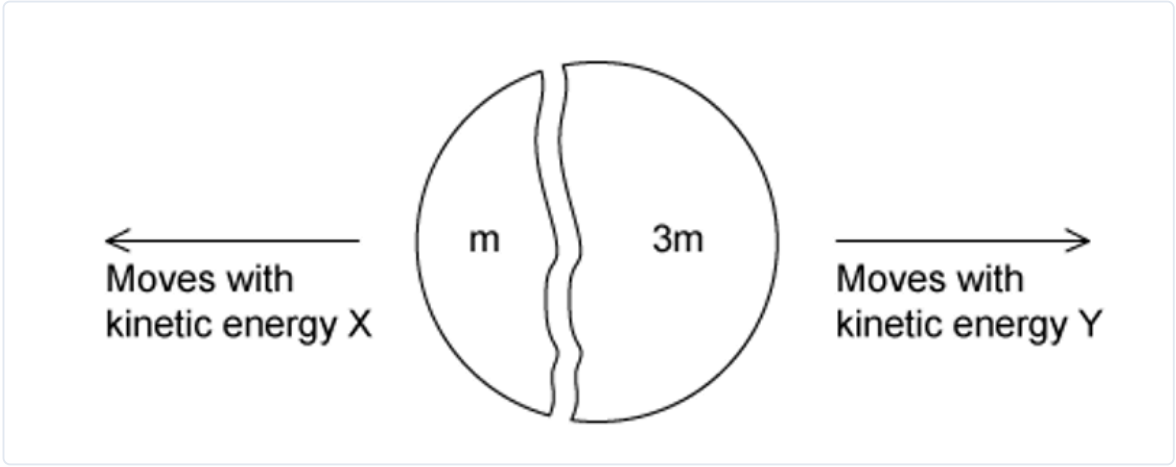
C mv^2

D $\frac{1}{2} mv^2$

Question 27 Hard [1 mark]

A stationary body explodes into two components of masses *m* and 3 *m*.

The components gain kinetic energies X and Y respectively.



What is the value of the ratio $\frac{X}{Y}$?

A $\frac{6}{1}$

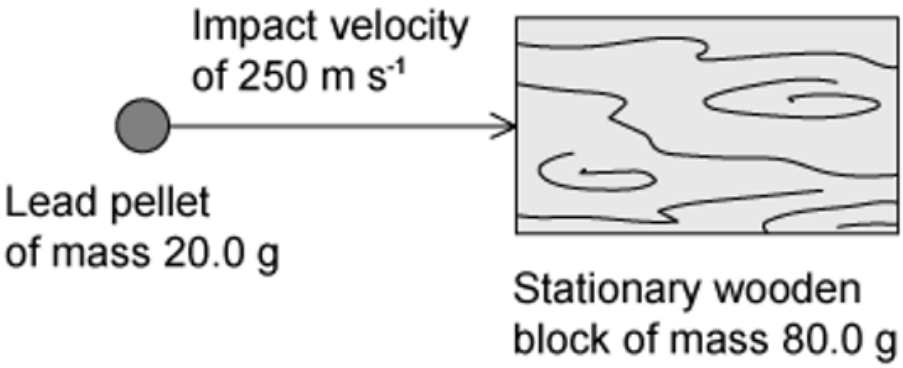
B $\frac{3}{1}$

C $\frac{1}{3}$

D $\frac{1}{6}$

Question 28 Hard [1 mark]

A lead pellet of mass 20.0 g is shot horizontally into a stationary wooden block of mass 80.0 g. The pellet hits the block with an impact velocity of 250 m s^{-1}. It embeds itself in the block and it does not emerge.



After the impact, the block and the pellet slide along a rough horizontal surface and come to rest in 8.4 s.

How far do the block and the pellet travel before coming to rest?

A 175 m

B 210 m

C 260 m

D 525 m

Question 29 Hard [1 mark]

A stationary nucleus has nucleon number *A*.

The nucleus decays by emitting a proton with speed *v* to form a new nucleus with speed *u*. The new nucleus and the proton move away from one another in opposite directions.

Which equation gives in terms of *A* and *u*?

A $v = \frac{A}{4} - 1 \quad u$

B $v = A - 1 \quad u$

C $v = Au$

D $v = A + 1 \quad u$

Question 30 Hard [1 mark]

A particle X has speed *v* and collides with a stationary identical particle Y. The collision is perfectly elastic.

What are the speed and direction of motion of each of the two particles after the collision?

	X	Y
A	stationary	v to the right
B	$\frac{v}{2}$ to the right	$\frac{v}{2}$ to the right
C	$\frac{v}{2}$ to the left	$\frac{v}{2}$ to the right
D	*v* to the left	stationary

A

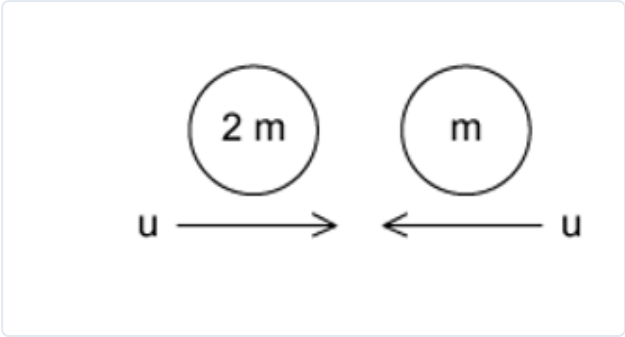
B

C

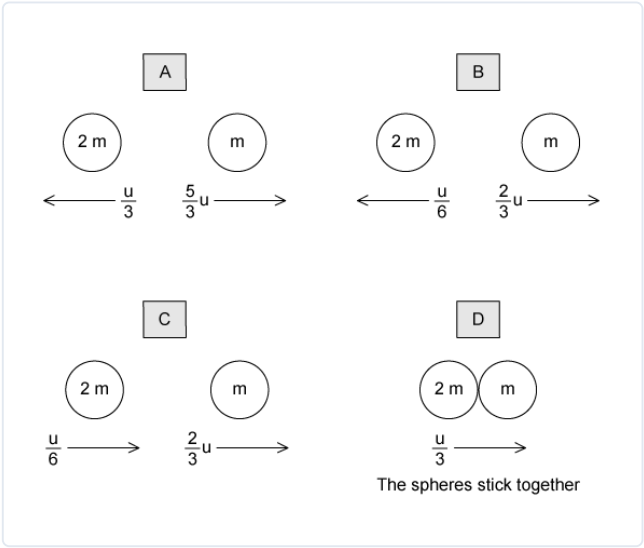
D

Question 31 Hard [1 mark]

The diagram shows two spherical masses approaching each other head-on at an equal speed *u*. One is of mass *m* and the other of mass 2 *m*.



Which diagram, showing the situation after the collision, is not consistent with the principle of conservation of momentum?



A

B

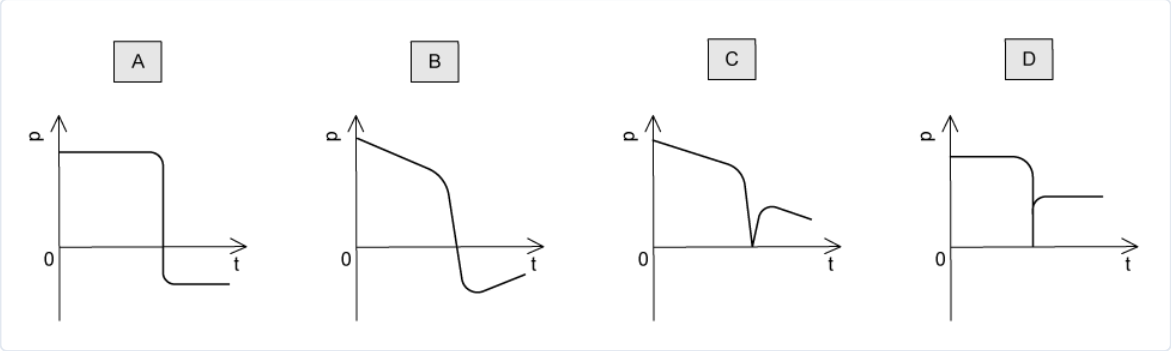
C

D

Question 32 Hard [1 mark]

An ice-hockey puck slides along a horizontal, frictionless ice-rink surface. It collides inelastically with a wall at right angles to its path, and then rebounds along its original path.

Which graph shows the variation with time *t* of the momentum *p* of the puck?



A

B

C

D

Question 33 Hard [1 mark]

A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speeds v_1 and v_2 respectively.

What is the ratio $\frac{v_1}{v_2}$?

A $\frac{M_2}{M_1}$

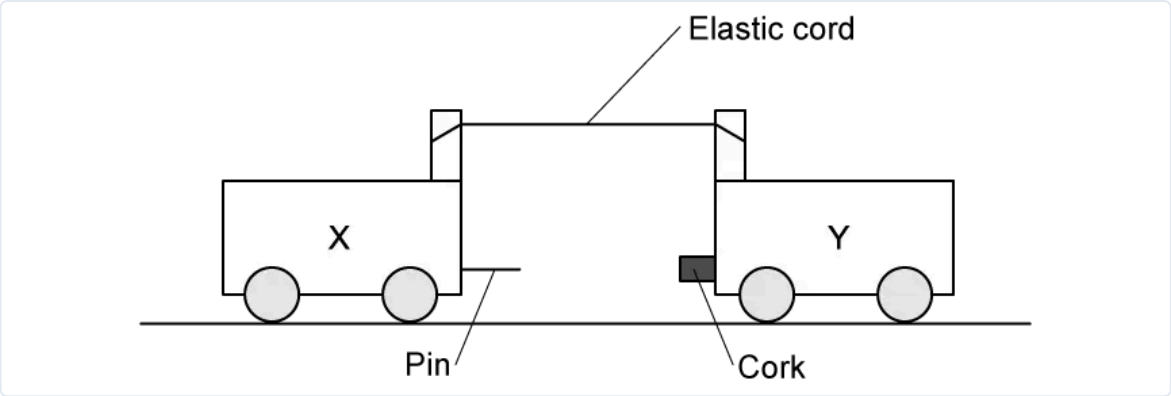
B $\frac{M_1}{M_2}$

C $\sqrt{\frac{M_2}{M_1}}$

D $\sqrt{\frac{M_1}{M_2}}$

Question 34 Hard [1 mark]

Two trolleys, X and Y, are connected by an elastic cord, of negligible mass, as shown below. Trolley X is twice the mass of trolley Y.



The two trolleys are pulled apart so that the elastic cord is stretched. The trolleys are then released at the same time and move towards each other. The trolleys stick together on impact.

Just before the trolleys collide, the speed of trolley X is 20 cm s^{-1}.

What is the speed of the trolleys after they collide?

A 0 cm s^{-1}

B 10 cm s^{-1}

C 20 cm s^{-1}

D 30 cm s^{-1}